1. (1) 當 $R=10k\Omega$ 時， $V_D = V_{DD} - I_0 R = 2.5V - 50\mu A \times 10k\Omega = 2V$

$\Rightarrow V_{GD} = V_{in} - V_D = 0V < V_T$ ， M_1 工作在飽和區

$I_0 = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(V_{GS} - V_T)^2 = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(V_{GT} - V_S)^2$

$\Rightarrow V_S = V_{in} - V_T - \sqrt{\frac{2I_0}{k_n'}\left(\frac{L}{W}\right)} = 1.30V$

(2) 當 $R=30k\Omega$ 時， $V_D = V_{DD} - I_0 R = 2.5V - 50\mu A \times 30k\Omega = 1V$

$\Rightarrow V_{GD} = V_{in} - V_D = 2V - 1V = 1V > V_T$， M_1 不工作在飽和區

若 M_1 工作區在速度飽和區，則有 $V_{GS} > V_T$ 且 $V_{DS} > V_{DSAT}$

$\Leftrightarrow$ ~~時~~ $V_S < \min(V_{in} - V_T, V_D - V_{DSAT}) = 0.4V$

顯然 $V_S = 0.4V$ 便是 M_1 的線性區和速度飽和區的分界

此時 $I_D = k_n' \left(\frac{W}{L}\right) V_{DSAT} \left(V_{in} - V_S - V_T - 0.5V_{DSAT}\right) = 594\mu A \neq I_0$，矛盾

所以 M_1 工作在線性區

$I_0 = k_n' \left(\frac{W}{L}\right)(V_D - V_S)\left[V_{in} - V_S - V_T - 0.5(V_D - V_S)\right]$

$50 = 1100(1 - V_S)(1.1 - 0.5V_S)$

$5 = 55V_S^2 - 176V_S + 121 \Leftrightarrow 55V_S^2 - 176V_S + 116 = 0$

$V_S = \frac{176 \pm \sqrt{176^2 - 4\times 55\times 116}}{110} = \frac{88 \pm \sqrt{88^2 - 55\times 116}}{55} = \frac{88 - \sqrt{88^2 - 55\times 116}}{55} = 0.93V$

(3) 當 $R=10k\Omega$ 且 $\lambda \neq 0$ 時，

$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(V_{in} - V_T - V_S)\left[1 + \lambda(V_D - V_S)\right]$

若 V_S 下降，則 I_D 上升，比 I_0 大，矛盾.

所以 V_S 只能升高

2、(1) i. 若M₁工作在饱和区，则 $V_i - V_o > V_T(V_o) = V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|}\right)$

$$= V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_o|} - \sqrt{2|\phi_F|}\right)$$

$$V_i - V_{DD} < V_T(V_o) \Leftrightarrow \cancel{饱}$$

$$\cancel{\Rightarrow V_o + V_{T0} + \gamma(\sqrt{\ }}$$

$$\cancel{\Rightarrow V_{DD} + V_T \quad V_{T0} + V_o + \gamma(\sqrt{\ }}$$

设 $V_{Def} = V_{T0} - \gamma\sqrt{2|\phi_F|} + \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)}$

$$\Rightarrow V_o + V_T(V_o) < V_i < V_{DD} + V_T(V_o)$$

$$I_o = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(V_i - V_o - V_T(V_o)\right)^2 \text{ ------ } (1)$$

ii. 若M₁工作在线性区，则 $\cancel{V_i > V_o + V_T(V_o), \; V_i}$ $V_i > \max(V_o, V_{DD}) + V_T(V_o)$

$$I_o = k_n'\left(\frac{W}{L}\right)(V_{DD} - V_o)\left[V_{in} - V_o - V_T(V_o) - 0.5(V_{DD} - V_o)\right] \text{ --- } (2)$$

(1)式：$I_o = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[V_i - V_o - V_{T0} - \gamma\left(\sqrt{|-2\phi_F + V_o|} - \sqrt{2|\phi_F|}\right)\right]^2$

$$\Leftrightarrow V_o + \gamma\sqrt{|-2\phi_F + V_o|} = V_i - V_{T0} + \gamma\sqrt{2|\phi_F|} - \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)} = V_i - V_{Def}$$

$$\Leftrightarrow \cancel{\gamma^2|-2\phi_F + V_o| = (V_i - V_{Def})^2 - 2(V_i - V_{Def})V_o + V_o^2}$$

$$\gamma^4(V_o - 2\phi_F)^2 = \left[V_o - (V_i - V_{Def})\right]^4 \qquad \boxed{V_i = V_o + \gamma\sqrt{|-2\phi_F + V_o|} + V_{Def}} \; (1^*)$$

$$\Leftrightarrow \left[V_o^2 - (2V_i - 2V_{Def} + \gamma^2)V_o + (V_i - V_{Def})^2 + 2\gamma^2\phi_F\right]\left[V_o^2 - (2V_i - 2V_{Def} - \gamma^2)V_o + (V_i - V_{Def})^2 - 2\gamma^2\phi_F\right] =$$

$$V_o = \frac{2V_i - 2V_{Def} + \gamma^2 \pm \sqrt{(2V_i - 2V_{Def} + \gamma^2)^2 - 4\left[(V_i - V_{Def})^2 + 2\gamma^2\phi_F\right]}}{2} \text{ 或 } \frac{2V_i - 2V_{Def} - \gamma^2 \pm \sqrt{(2V_i - 2V_{Def} - \gamma^2)^2 - 4\left[(V_i - V_{Def})^2 - 2\gamma^2\phi_F\right]}}{2}$$

$$= \frac{2V_i - 2V_{Def} + \gamma^2 \pm \sqrt{\gamma^2(4V_i - 4V_{Def} + \gamma^2) - 8\gamma^2\phi_F}}{2} \text{ 或 } \frac{2V_i - 2V_{Def} - \gamma^2 \pm \sqrt{(\gamma^2 + 4V_{Def} - 4V_i)\gamma^2 + 8\gamma^2\phi_F}}{2}$$

$$= \boxed{\frac{2V_i - 2V_{Def} + \gamma^2 \pm \gamma\sqrt{4V_i - 4V_{Def} + \gamma^2 - 8\phi_F}}{2}} \text{ 或 } \frac{2V_i - 2V_{Def} - \gamma^2 \pm \gamma\sqrt{8\phi_F - (4V_i - 4V_{Def} - \gamma^2)}}{2} \text{ (舍去)}$$

~~ii. 若M₁工作线性区~~

(2)式：$I_o = k_n'\left(\frac{W}{L}\right)(V_{DD} - V_o)\left(V_{in} - V_o - V_{T0} - \gamma\left(\sqrt{|-2\phi_F + V_o|} - \sqrt{|2\phi_F|}\right) - 0.5V_{DD} + 0.5V_o\right)$

$$V_i = \frac{\frac{I_o}{k_n'}\left(\frac{L}{W}\right)}{V_{DD} - V_o} + 0.5V_o + V_{T0} + 0.5V_{DD} + \gamma\left(\sqrt{|-2\phi_F + V_o|} - \sqrt{|2\phi_F|}\right) \qquad (2^*)$$

(1*)和(2*)是 V_i 与 V_o 的函数关系.

$$V_i = \begin{cases} V_o + \gamma\sqrt{|1-2\phi_F+V_o|} + V_{T0} - \gamma\sqrt{2|\phi_F|} + \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)}, & V_o < V_{DD} - \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)} = 2.13\,V \\ \dfrac{\frac{I_o}{k_n'}\left(\frac{L}{W}\right)}{V_{DD}-V_o} + 0.5(V_{DD}+V_o) + V_{T0} + \gamma\left(\sqrt{|1-2\phi_F+V_o|} - \sqrt{2|\phi_F|}\right), & V_o > 2.13\,V \end{cases}$$

$$V_T = V_{T0} + \gamma\left(\sqrt{|1-2\phi_F+V_o|} - \sqrt{2|\phi_F|}\right) = V_{T0} + \gamma\left(\sqrt{V_o-2\phi_F} - \sqrt{-2\phi_F}\right)$$

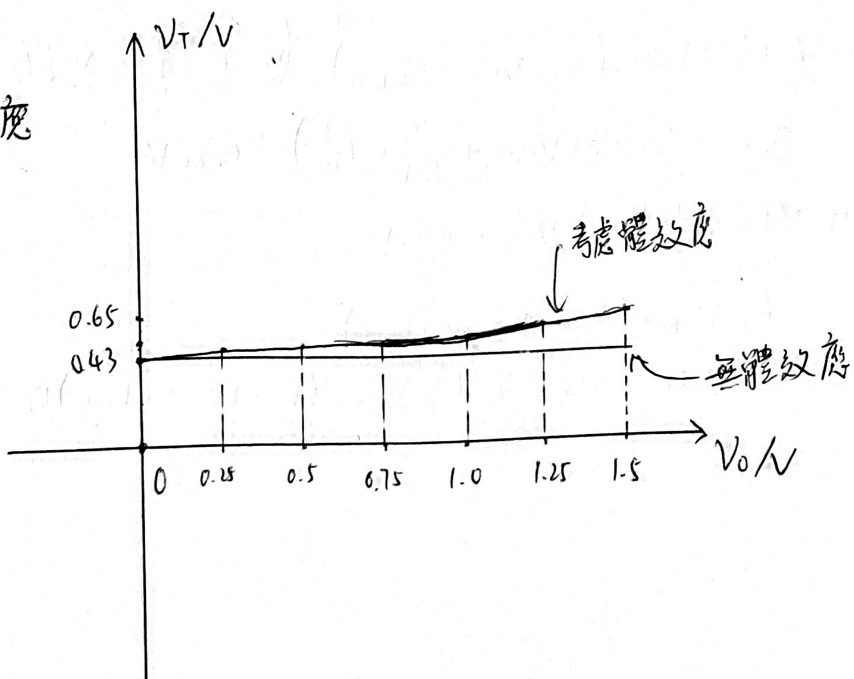
$$\Rightarrow V_o = 2\phi_F + \left(\frac{V_T(V_o)-V_{T0}}{\gamma} + \sqrt{2|\phi_F|}\right)^2$$

$$\Rightarrow \cancel{V_i} \quad V_i = \begin{cases} 2\phi_F + \left(\frac{V_T(V_o)-V_{T0}}{\gamma} + \sqrt{2|\phi_F|}\right)^2 + V_T(V_o) + \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)}, & V_o < 2.13\,V \\ \dfrac{\frac{I_o}{k_n'}\left(\frac{L}{W}\right)}{V_{DD}-2\phi_F-\left(\frac{V_T(V_o)-V_{T0}}{\gamma}+\sqrt{2|\phi_F|}\right)^2} + 0.5\left(V_{DD}+2\phi_F+\left(\frac{V_T(V_o)-V_{T0}}{\gamma}+\sqrt{2|\phi_F|}\right)^2\right) + V_T(V_o), & V_o \geq 2.13\,V \end{cases}$$

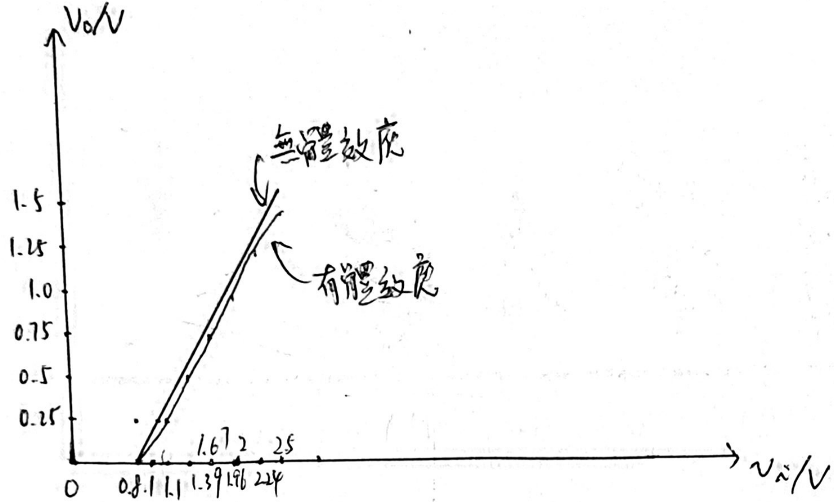
若不考慮體效應，轉換點定值為 $V_{T0} + \sqrt{\frac{2I_o}{k_n'}\left(\frac{L}{W}\right)}$
$= 0.80605\,V$

~~(2) $V_T = V_{T0} + \gamma\left(\sqrt{|1-2\phi_F+V_o|} - \sqrt{2|\phi_F|}\right)$ 考~~

$$(2)\quad V_T = \begin{cases} V_{T0} + \gamma\left(\sqrt{|1-2\phi_F+V_o|} - \sqrt{2|\phi_F|}\right), & \text{考慮體效應} \\ V_{T0}, & \text{不考慮體效應} \end{cases}$$



(3)



當 $V_o = 1.5V$ 時，考慮體效應 $V_i = 2.5251V$

不考慮體效應 $V_i = 1.5V + 0.80605V = 2.30605V$

誤差：$0.21905V$

3、當 $V_{in} < V_T = 0.4V$ 時，M_1 工作在截止區，$V_{out} = V_{DD} = 2.5V$

當 $V_{in} > 0.4V$ 時，

(i) 當 M_1 工作在飽和區，$V_{out} > V_{in} - V_T$

$$V_{out} = V_{DD} - I_D R_L = V_{DD} - \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{in}-V_T)^2(1+\lambda V_{out})$$

$$\Leftrightarrow V_{out} = \frac{V_{DD}+\frac{1}{\lambda}}{1+\frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{in}-V_T)^2\lambda R_L} - \frac{1}{\lambda} = \frac{8.05}{1+0.68(V_{in}-0.4)^2} - 6.25\,(V)$$

(ii) 當 M_1 工作在速度飽和區，$V_{DSAT} < V_{out} < V_{in} - V_T$

$$V_{out} = V_{DD} - I_D R_L = V_{DD} - k_n'\left(\frac{W}{L}\right)V_{DSAT}\left(V_{in}-V_T-0.5V_{DSAT}\right)R_L(1+\lambda V_{out})$$

$$\cancel{= 2.5 - 3.825(V_{in}-0.625)\,V} = 2.5 - 3.825(V_{in}-0.625)(1+\lambda V_{out})$$

$$V_{out} = \frac{V_{DD}+\frac{1}{\lambda}}{1+k_n'\left(\frac{W}{L}\right)V_{DSAT}(V_{in}-V_T-0.5V_{DSAT})\lambda R_L} - \frac{1}{\lambda} = \frac{8.05}{1+0.612(V_{in}-0.625)} - 6.25\,(V)$$

(iii) 當 M_1 工作在線性區，$V_{out} < V_{DSAT}$

$$V_{out} = V_{DD} - k_n'\left(\frac{W}{L}\right)V_{out}(V_{in}-V_T-0.5V_{out})(1+\lambda V_{out})R_L$$

飽和區與速度飽和區分界線：$0.68(V_{in}-0.4)^2 = 0.612(V_{in}-0.625)$

$$\cancel{0.068V_{in}^2} \quad 10V_{in}^2 - 8V_{in} + 1.6 = 9V_{in} - 5.625$$

$$10V_{in}^2 - 17V_{in} + 7.225 = 0$$

$$100V_{in}^2 - 170V_{in} + 72.25 = 0$$

$$(10V_{in} - 8.5)^2 = 0 \Leftrightarrow V_{in} = 0.85V$$

速度飽和區與線性區分界線

$$\frac{8.05}{1+0.612(V_{in}-0.625)} - 6.25 = 0.45$$

$$\frac{8.05}{6.70} - 1 = 0.612(V_{in}-0.625)$$

$$\Rightarrow V_{in} = \cancel{1.16V}\ 0.95V$$